Section 2.8 — Rational Functions

Math 103: College Algebra

1 Identifying Rational Expressions

Identify each expression as rational or not.

1. $\dfrac{2x^2 - 5x + 3}{24x^3 - 5}$
2. $\dfrac{wx^2y}{\sqrt{25x}}$
3. $\dfrac{2x - 4}{x + \sqrt{2}}$
4. $2x^2 - 4x + 5z$

2 Negative Exponents

Rewrite each expression involving negative exponents as a rational expression.

5. $2x^{-4}y^2z^{-1}$
6. $\frac{2}{3}u^{-2}w^5$
7. $12a^{-1}b^{-7}$
8. $\frac{1}{4}j^3k^{-3}$

3 Rational Functions

For each pair of functions f and g, write the function $\left(\frac{f}{g}\right)(x)$ as a rational function, and find its domain. **Do not simplify** your rational function.

9. $f(x) = 2x^2 - 5x + 3, \quad g(x) = 2x^2 - 5x - 7$
10. $f(x) = 2x - 3, \quad g(x) = 4x^2 - 9$

4 Evaluating Rational Functions

Let $h(x) = \dfrac{2x^2 - 5x + 3}{x^3 - 3x^2 + 3x - 3}$. Find

11. $h(2)$
12. $h(-3)$
13. $h(\frac{1}{2})$
14. $h(1)$

Let $s(x) = \dfrac{x^2 - 5x - 11}{2x^2 + 5}$

15. $s(7)$
16. $s(-1)$
17. $s(1.5)$
18. $s(0)$

5 Simplifying Expressions

Simplify each expression. You may leave your answer in factored form.

19. $\dfrac{14h^2k}{21h^3}$
20. $-\dfrac{5a^3b}{a^2b^4}$
21. $-\dfrac{5x^2y}{10x^3y}$
22. $\dfrac{2x^2yu^4}{8x^7u^8z^2}$
23. $\dfrac{3(m-2)}{5(m-2)}$
24. $\dfrac{(x+4)(3x-5)(x^2+1)}{(x-4)(3x-5)(x+1)^2}$
25. $\dfrac{5-b}{b-5}$
26. $\dfrac{10x - 20}{15x - 30}$
27. $\dfrac{3x - 5}{10x - 6x^2}$
28. $\dfrac{6b^2 + b - 12}{10b^2 + 13b - 3}$
29. $\dfrac{px + py + qx + qy}{mx - nx + my - ny}$
30. $\dfrac{a^2 - b^2}{a - b}$
31. $\dfrac{x^2 - y^2}{x^3 - y^3}$
32. $\dfrac{x^3 - 16x}{x^3 + 6x^2 + 8x}$
33. $\dfrac{8n^3 + 20n^2 - 20n - 50}{4n^2 + 12n + 5}$
34. $\dfrac{x^3 - 8}{x^3 + 2x^2 + 4x}$

6 Simplifying Functions

Simplify each function. Note the domain restrictions, if any. You may leave your answer in factored form.

35. $f(x) = \dfrac{3x}{3x^2 - 9x}$
36. $g(x) = \dfrac{x - 1}{4x^2 + 4x - 8}$
37. $h(x) = \dfrac{3x - 7}{3x^2 - 4x - 7}$
38. $k(x) = \dfrac{x^2 - 9}{2x^2 - 7x + 3}$
39. $s(x) = \dfrac{4x^2 + 5x - 6}{x^3 + 8}$
40. $t(x) = \dfrac{2x^2 + 7x + 6}{18 + 9x - 8x^2 - 4x^3}$